Please write clearly in block capitals.

Centre number Candidate number

Surname ______________________________________________

Forename(s) ___________________________________________

Candidate signature ______________________________________

A-level

# ECONOMICS

## Paper 3 Economic principles and issues

Predicted Paper **Time allowed: 2 hours**

### Materials

For this paper you must have:

- an AQA 16-page answer book
- a calculator.

### Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Write the information required on the front of your answer book. The Paper Reference is 7136/3.
- Answer **ALL** questions.
- In Section A, for each multiple-choice question, select one answer from A, B, C or D.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.
- There are 30 marks for Section A and 50 marks for Section B.
- You will be marked on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

### Advice

- You are advised to spend 40 minutes on Section A and 80 minutes on Section B.

**Predicted Paper — For revision purposes only [Set B — Hard]**

## Section A

**Answer all questions. Each question is worth 1 mark.**

**For each question, select the single best option: A, B, C or D.**

**Question 01** **[1 mark]**

The law of diminishing marginal returns applies in which one of the following cases?

**A** In the long run, when all factors are variable.
**B** In the short run, when at least one factor is fixed.
**C** Whenever there are constant returns to scale.
**D** Only in agricultural production.

**Question 02** **[1 mark]**

A firm maximises revenue when:

**A** Price is at its maximum.
**B** Marginal revenue equals marginal cost.
**C** Marginal revenue equals zero.
**D** Average revenue equals average cost.

**Question 03** **[1 mark]**

Which one of the following is the most likely consequence of a rise in the long-run trend growth of labour productivity?

**A** A leftward shift in long-run aggregate supply.
**B** A rise in the natural rate of unemployment.
**C** A rightward shift in long-run aggregate supply.
**D** A rise in demand-pull inflation.

**Question 04** **[1 mark]**

In a contestable market, the most important determinant of incumbent firms' behaviour is:

**A** The number of firms currently in the market.
**B** The level of fixed costs in the industry.
**C** The threat of potential new entrants.
**D** Government subsidies to existing firms.

**Question 05** **[1 mark]**

A monopsonist in a labour market is most likely to:

**A** Pay a wage equal to the marginal revenue product of labour.
**B** Pay a wage above the competitive equilibrium wage.
**C** Pay a wage below the marginal revenue product of labour.
**D** Employ more workers than would exist under perfect competition.

**Question 06** **[1 mark]**

An appreciation of a country's currency is most likely to:

**A** Reduce cost-push inflationary pressure by making imports cheaper in domestic currency.
**B** Reduce the foreign-currency price of exports.
**C** Improve the current account of the balance of payments in the short run.
**D** Encourage tourism to that country.

**Question 07** **[1 mark]**

A fall in the national savings rate, with investment unchanged, will most likely result in:

**A** An improvement in the current account balance.
**B** A deterioration in the current account balance.
**C** A balanced current account.
**D** A reduction in the capital account balance.

**Question 08** **[1 mark]**

If the Marshall-Lerner condition holds, a currency depreciation will:

**A** Always worsen the trade balance.
**B** Always leave the trade balance unchanged.
**C** Improve the trade balance once prices and quantities have fully adjusted.
**D** Reduce the sensitivity of trade flows to relative prices.

**Question 09** **[1 mark]**

Supply-side policies that increase the flexibility of labour markets are most likely to:

**A** Reduce the natural rate of unemployment.
**B** Increase cyclical unemployment.
**C** Reduce productivity growth.
**D** Raise the equilibrium real wage for unskilled workers.

**Question 10** **[1 mark]**

Regulatory capture most accurately describes the situation in which:

**A** Regulators protect consumers against producer exploitation.
**B** Regulators come to act primarily in the interests of the industries they regulate.
**C** Regulators set prices at the allocatively efficient level.
**D** Regulators merge to form a single super-regulator.

**Question 11** **[1 mark]**

The table below shows output and costs for a firm at different employment levels, holding capital fixed.

**Table 1**

| Workers | Total output | Total cost (£) | Marginal product |
|---|---|---|---|
| 1 | 10 | 50 | — |
| 2 | 22 | 80 | 12 |
| 3 | 38 | 110 | 16 |
| 4 | 50 | 140 | 12 |
| 5 | 58 | 170 | 8 |
| 6 | 62 | 200 | 4 |

Based on Table 1, diminishing marginal returns to labour begin from the employment of which worker?

**A** The second worker.
**B** The third worker.
**C** The fourth worker.
**D** The sixth worker.

**Question 12** **[1 mark]**

The table below shows the total revenue and total cost for a perfectly competitive firm.

**Table 2**

| Output (units) | Price (£) | Total Revenue (£) | Total Cost (£) |
|---|---|---|---|
| 0 | 10 | 0 | 20 |
| 10 | 10 | 100 | 60 |
| 20 | 10 | 200 | 110 |
| 30 | 10 | 300 | 180 |
| 40 | 10 | 400 | 280 |
| 50 | 10 | 500 | 420 |

Based on Table 2, the firm's profit-maximising output and supernormal profit are:

**A** 30 units; £120 supernormal profit.
**B** 40 units; £120 supernormal profit.
**C** 40 units; £40 supernormal profit.
**D** 50 units; £80 supernormal profit.

**Question 13** **[1 mark]**

The table below shows quarterly real GDP growth (annualised) for a country.

**Table 3**

| Quarter | Real GDP growth (%) |
|---|---|
| Q1 2023 | +0.4 |
| Q2 2023 | +0.2 |
| Q3 2023 | –0.1 |
| Q4 2023 | –0.3 |
| Q1 2024 | +0.1 |
| Q2 2024 | +0.4 |

Based on Table 3, the country is most accurately described as having experienced:

**A** A deep recession throughout 2023.
**B** A technical recession spanning Q3 and Q4 2023.
**C** An economic boom in 2024.
**D** Stagflation throughout the period.

**Question 14** **[1 mark]**

The table below shows data for two small open economies, P and Q.

**Table 4**

| Indicator | Economy P | Economy Q |
|---|---|---|
| GDP growth (%) | 3.2 | 3.0 |
| Inflation (%) | 4.1 | 1.8 |
| Bank policy rate (%) | 2.5 | 2.5 |
| Current account (% GDP) | –4.8 | +1.2 |
| Exchange rate regime | Fixed | Floating |

Based on Table 4, economy P is most likely to face:

**A** A loss of competitiveness and pressure on the fixed exchange rate.
**B** An appreciation of its fixed currency.
**C** A rapid reduction in imports.
**D** Deflationary pressure in the short run.

**Question 15** **[1 mark]**

The table below shows the budget position and output gap estimates for Country Z.

**Table 5**

| Year | Output gap (%) | Budget balance (% GDP) |
|---|---|---|
| Year 1 | +1.5 (positive) | –2.0 (deficit) |
| Year 2 | +0.5 | –2.5 |
| Year 3 | –1.8 (negative) | –4.5 |

Based on Table 5, the fiscal policy stance can best be described as:

**A** Strongly counter-cyclical in Year 1.

**B** Pro-cyclical in Year 1 and counter-cyclical in Year 3.

**C** Pro-cyclical throughout the period.

**D** Counter-cyclical throughout the period.

**Question 16** **[1 mark]**

The table below shows the marginal social costs (MSC) and benefits (MSB) of emissions abatement.

**Table 6**

| Abatement (% below baseline) | MSC of abatement (£) | MSB of abatement (£) |
|---|---|---|
| 20% | 20 | 80 |
| 40% | 40 | 60 |
| 60% | 60 | 60 |
| 80% | 80 | 40 |

Based on Table 6, the allocatively efficient level of abatement is:

**A** 20%.
**B** 40%.
**C** 60%.
**D** 80%.

**Question 17** **[1 mark]**

The table below shows the demand schedule faced by a monopolist and the firm's marginal cost.

**Table 7**

| Q | P (£) | TR (£) | MR (£) | MC (£) |
|---|---|---|---|---|
| 1 | 14 | 14 | 14 | 2 |
| 2 | 12 | 24 | 10 | 4 |
| 3 | 10 | 30 | 6 | 6 |
| 4 | 8 | 32 | 2 | 8 |
| 5 | 6 | 30 | –2 | 10 |

Based on Table 7, the profit-maximising monopolist's price and deadweight loss relative to allocative efficiency are:

**A** P = £14; there is no deadweight loss.
**B** P = £10; there is a deadweight loss because P > MC at Q = 3.
**C** P = £8; there is no deadweight loss.
**D** P = £6; the monopoly is allocatively efficient.

**Question 18** **[1 mark]**

The table below shows the Gini coefficients of five countries in 2010 and 2023.

**Table 8**

| Country | Gini 2010 | Gini 2023 |
|---|---|---|
| A | 0.28 | 0.32 |
| B | 0.38 | 0.36 |
| C | 0.45 | 0.48 |
| D | 0.32 | 0.28 |
| E | 0.40 | 0.40 |

Based on Table 8, which country has experienced the largest reduction in income inequality over the period?

**A** Country A.

**B** Country B.

**C** Country C.

**D** Country D.

**Question 19** **[1 mark]**

The table below shows the labour productivity (output per worker, £000s per year) in two regions of an economy.

**Table 9**

| Region | 2014 | 2024 |
|---|---|---|
| North | 41 | 44 |
| South | 52 | 68 |

Based on Table 9, which one of the following can be correctly concluded?

**A** Productivity convergence has occurred between the two regions.

**B** The absolute productivity gap between South and North has narrowed.

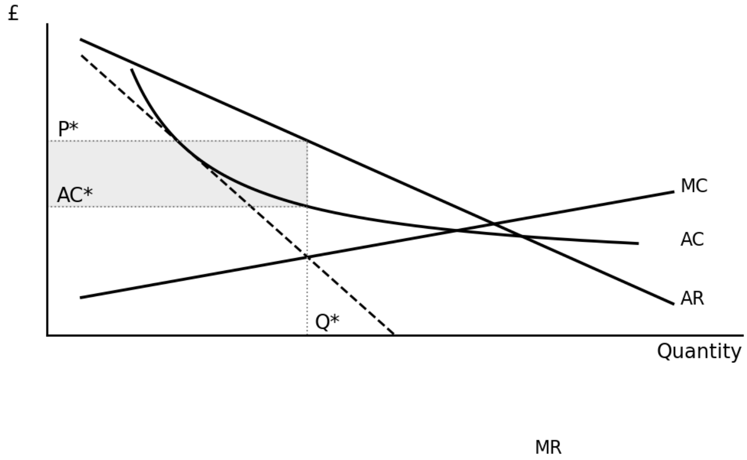
**C** The absolute productivity gap between South and North has widened.

**D** The North experienced faster productivity growth than the South.

**Question 20** **[1 mark]**

Figure 1 shows the cost and revenue curves for a profit-maximising monopoly.

**Figure 1 — Monopoly profit maximisation**




Based on Figure 1, the monopolist's supernormal profit is represented by:

**A** The whole area under the AR curve up to Q*.

**B** The rectangle bounded by (P* − AC*) × Q*.

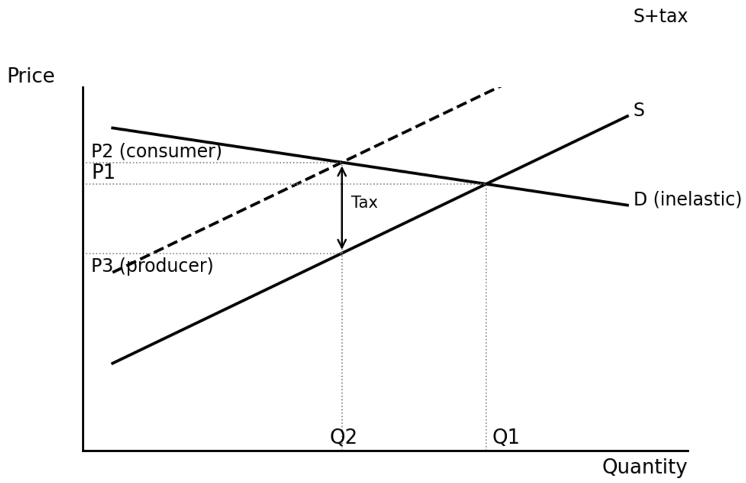
**C** The triangle between MR, MC and the vertical axis.

**D** The area between the AC and MC curves.

**Question 21** **[1 mark]**

Figure 2 shows the incidence of an indirect tax on a good with relatively inelastic demand.

**Figure 2 — Indirect tax incidence**




Based on Figure 2, which one of the following statements is correct?

**A** Most of the tax burden falls on producers because demand is inelastic.

**B** Most of the tax burden falls on consumers because demand is inelastic.

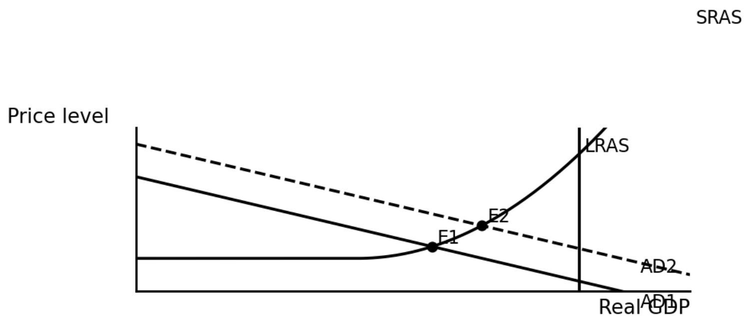
**C** The burden is split equally between consumers and producers.

**D** The incidence on consumers depends only on the elasticity of supply.

**Question 22** **[1 mark]**

Figure 3 shows an economy with spare capacity experiencing an expansionary demand shock.

**Figure 3 — AD shift with spare capacity**




Based on Figure 3, near LRAS the same rightward shift in AD would most likely result in:

**A** A smaller rise in the price level and a larger rise in real GDP than shown.

**B** A larger rise in the price level and a smaller rise in real GDP than shown.

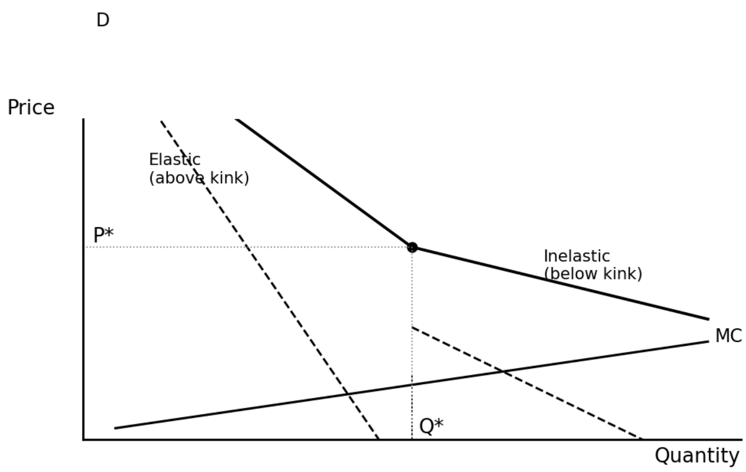
**C** No change in either price level or real GDP.

**D** A fall in both the price level and real GDP.

**Question 23** **[1 mark]**

Figure 4 shows the kinked demand curve model of oligopoly.

**Figure 4 — Kinked demand curve (oligopoly)**




Based on Figure 4, the discontinuity (vertical gap) in the marginal revenue curve is most consistent with:

**A** Perfectly competitive behaviour.

**B** Price rigidity despite changes in marginal cost within a range.

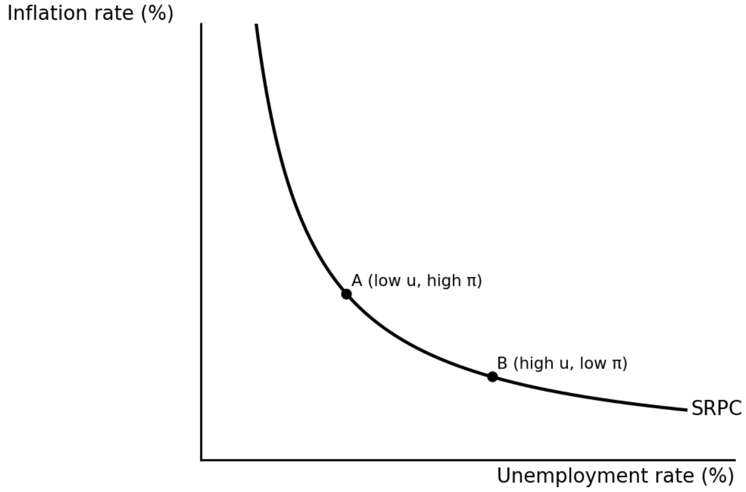
**C** Frequent price-cutting by all firms.

**D** A firm that faces a perfectly elastic demand curve.

**Question 24** **[1 mark]**

Figure 5 shows a short-run Phillips curve.

**Figure 5 — Short-run Phillips curve**




Based on Figure 5 and your knowledge of economics, in the long run the relationship is most likely to be:

**A** The same as the short-run relationship shown.

**B** Permanently vertical at the natural rate of unemployment.

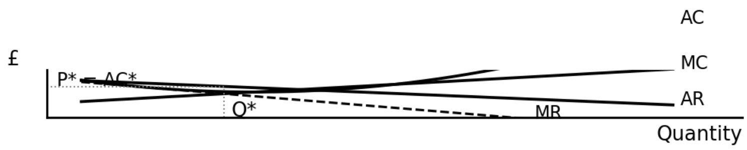
**C** Permanently upward-sloping.

**D** Horizontal at a constant rate of inflation.

**Question 25** **[1 mark]**

Figure 6 shows a firm in long-run equilibrium under monopolistic competition.

**Figure 6 — Monopolistic competition: long-run equilibrium**




Based on Figure 6, the firm operates with:

**A** Productive efficiency (minimum AC) but allocative inefficiency (P > MC).

**B** Allocative inefficiency (P > MC) AND productive inefficiency (P above minimum AC).

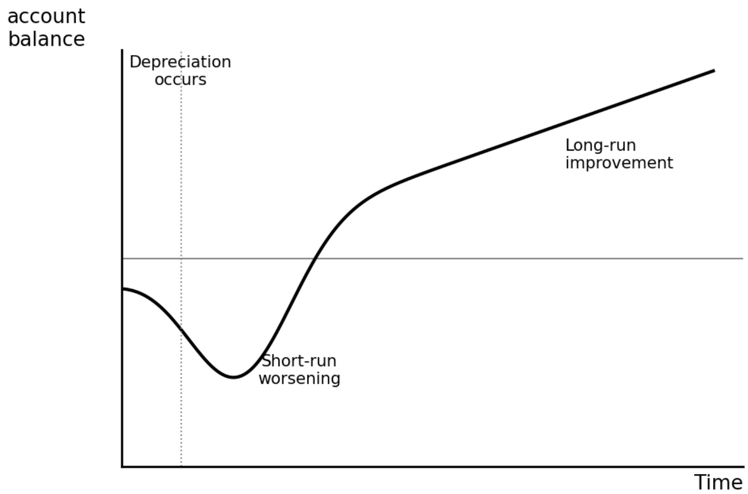
**C** Both productive and allocative efficiency.

**D** Supernormal profits in the long run.

**Question 26** **[1 mark]**

Figure 7 shows the J-curve effect after a depreciation of the currency.

**Figure 7 — J-curve after depreciation**




Based on Figure 7, the long-run improvement in the current account after depreciation depends most on:

**A** Capital flows reversing immediately.

**B** The Marshall-Lerner condition being met.

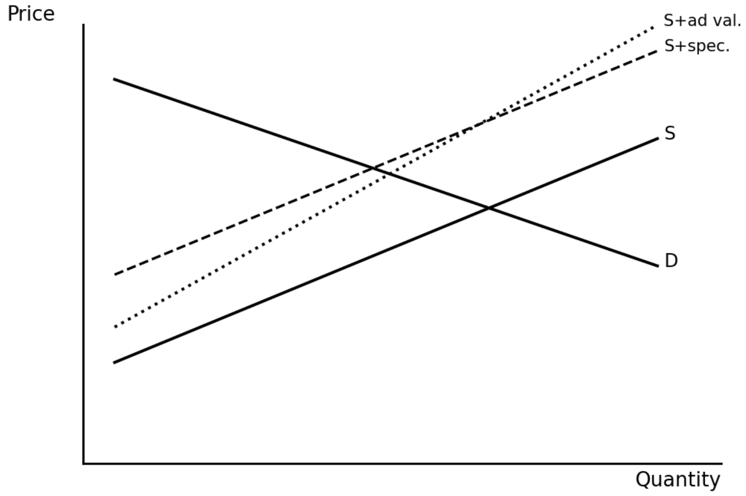
**C** Tariffs imposed by trading partners.

**D** A fall in aggregate demand.

**Question 27** **[1 mark]**

Figure 8 compares a specific (per-unit) tax with an ad valorem tax.

**Figure 8 — Specific tax vs ad valorem tax**




Based on Figure 8, an ad valorem tax:

**A** Raises the price of low-priced goods more than high-priced goods.

**B** Raises the price of high-priced goods more than low-priced goods.

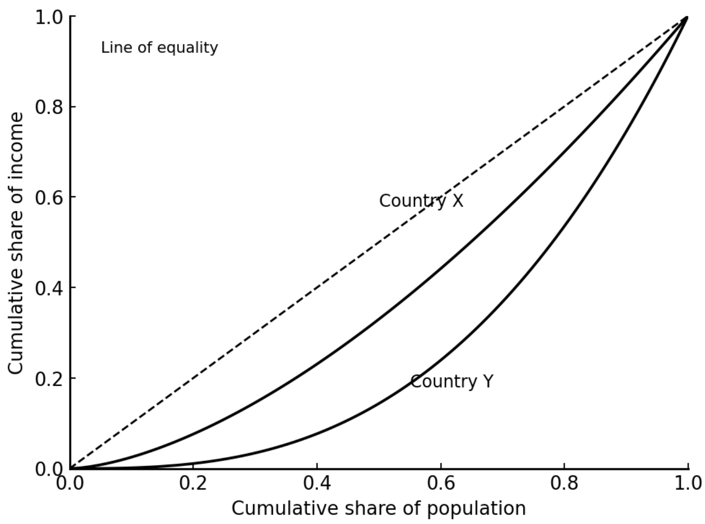
**C** Raises the price by the same absolute amount for all goods.

**D** Has no effect on consumer prices.

**Question 28** **[1 mark]**

Figure 9 shows Lorenz curves for two countries, X and Y.

**Figure 9 — Lorenz curves**




Based on Figure 9, the area between the line of equality and the Lorenz curve for Country Y relative to Country X indicates that:

**A** Country Y has lower income inequality than Country X.

**B** Country Y has higher income inequality than Country X.

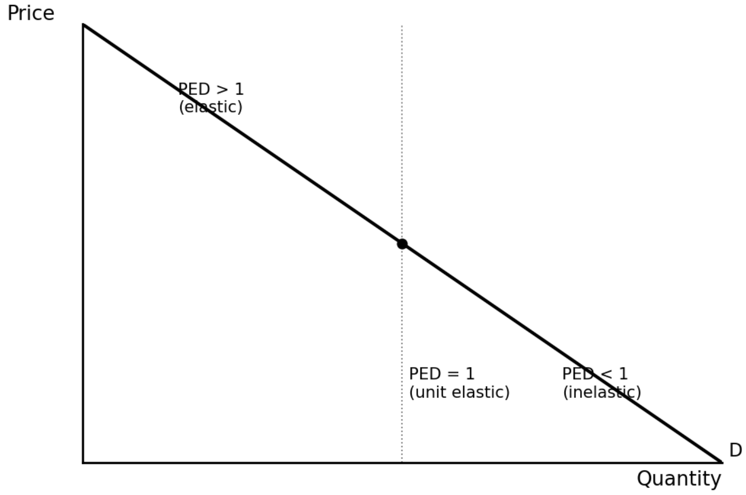
**C** The two countries have identical income distributions.

**D** Country Y has a lower Gini coefficient.

**Question 29** **[1 mark]**

Figure 10 shows a linear demand curve with regions of different elasticity.

**Figure 10 — PED along a linear demand curve**




Based on Figure 10, a firm currently pricing at the point where PED = 1 would find that a small rise or fall in price:

**A** Raises total revenue.

**B** Reduces total revenue.

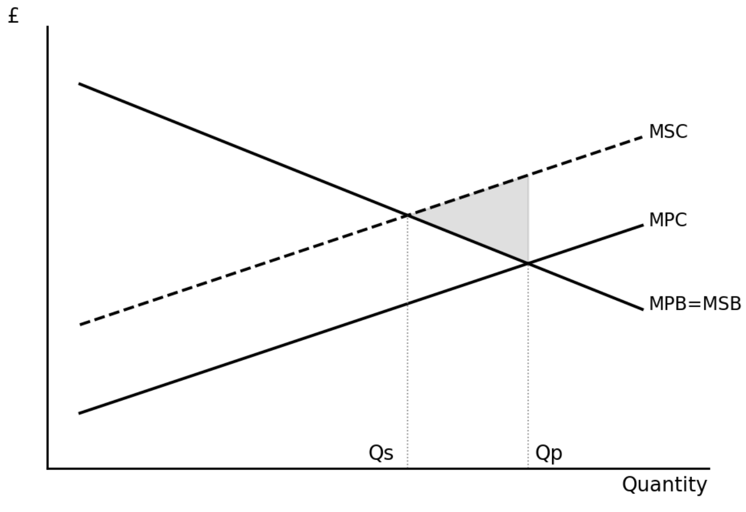
**C** Leaves total revenue approximately unchanged.

**D** Shifts the demand curve.

**Question 30** **[1 mark]**

Figure 11 shows a negative production externality.

**Figure 11 — Negative production externality**




Based on Figure 11, a Pigouvian tax set equal to the marginal external cost would:

**A** Raise output above Qp.

**B** Leave output unchanged at Qp.

**C** Reduce output from Qp to Qs and eliminate the deadweight loss.

**D** Reduce output to zero.

# Section B

**Answer all questions. Each question's marks are shown in brackets.**

**Study Extracts A, B, C and D, then answer Questions 31 to 33.**

## Case study: UK trade, Brexit and the services economy

### Extract A: UK trade composition, 2015–2024

**Figure 12: Selected UK trade and exchange-rate indicators**

| Indicator | 2015 | 2019 | 2022 | 2024 |
|---|---|---|---|---|
| Goods exports (£bn, annual) | 286 | 372 | 403 | 386 |
| Services exports (£bn, annual) | 229 | 306 | 408 | 452 |
| Goods trade balance (£bn) | –122 | –149 | –210 | –198 |
| Services trade balance (£bn) | +91 | +119 | +158 | +183 |
| Sterling ERI (2015 = 100) | 100.0 | 80.4 | 79.1 | 83.6 |
| Share of trade with EU (%) | 51.1 | 46.8 | 42.0 | 41.4 |

*Source: Office for National Statistics / Bank of England, 2024*

### Extract B: Brexit and UK goods trade

The UK left the EU single market on 31 January 2020, with new customs arrangements fully implemented by January 2021. Goods trade with the EU contracted sharply — a 2024 study by the Centre for European Reform estimated UK goods exports to the EU are around 15% below their counterfactual. Non-tariff barriers — customs declarations, rules-of-origin certificates and regulatory divergence — have raised transaction costs significantly, particularly for smaller firms. The Trade and Cooperation Agreement (TCA) signed in December 2020 provides tariff-free, quota-free access for goods meeting origin rules, but without the frictionless flow of the single market.

*Source: News reports, August 2024*

### Extract C: The rise of UK services exports

UK services exports have risen strongly since 2015, in contrast to the weaker goods performance. Financial services, legal services, consultancy, computer and information services, and creative industries are the UK's largest services export categories. The UK's services surplus has grown to £183bn in 2024 — the largest in the world in absolute terms. Non-EU markets have been particularly important: the UK's services exports to the US, China, India and the Gulf have grown faster than those to the EU.

Economists note that services exports have benefited from features that are less affected by Brexit: digital delivery reduces the importance of physical borders; services are often concentrated in London, where agglomeration effects are strong; and many services firms serve global (not just European) customers. However, services are governed by different trade agreements than goods — services trade in the TCA is significantly shallower than in the EU single market. A 2024 Resolution Foundation report warned that EU moves to reduce services dependence on the UK (e.g. requirements for financial services firms to have EU subsidiaries) could gradually erode this advantage.

*Source: News reports, November 2024*

### Extract D: The case for a services-led industrial strategy

Some economists argue that the UK should embrace its comparative advantage in services and design industrial policy around services-led growth. Specific proposals include: deeper services trade agreements with the US, India and major Asian economies; investment in the UK's "invisible infrastructure" (digital networks, regulatory frameworks, skills); deregulation of professional services export barriers; and support for high-growth technology start-ups via R&D; incentives and growth-capital support.

Critics argue that a services-led strategy risks three things. First, widening regional inequality: services jobs are concentrated in London and the South East, exacerbating the north-south divide. Second, volatility: financial and business services are sensitive to global shocks and regulatory change — the 2008 crisis hit the UK harder than peers because of the financial sector's dominance. Third, neglect of strategic industries: semiconductors, pharmaceuticals, clean energy manufacturing and defence equipment all have positive externalities and strategic significance that pure comparative-advantage analysis undervalues. These critics favour a balanced approach that combines support for services with targeted interventions to rebuild domestic manufacturing capabilities.

*Source: News reports, January 2025*

**3 1** Using the data in the Extracts, examine the extent to which the UK has experienced a structural shift from goods to services in its trade performance since 2015. **[10 marks]**

**3 2** Using Extracts A, B and C and your knowledge of economics, explain why the UK's comparative advantage may lie in services rather than in goods. **[15 marks]**

**3 3** Considering the information in Extract D and the original evidence in Extracts A, B and C, do you recommend that the UK government should adopt a services-led industrial strategy as the main approach to improving the UK's long-run trade performance? Justify your recommendation. **[25 marks]**

## END OF QUESTIONS